

Maths-1

$$f: \mathbb{Z} \rightarrow \mathbb{Z}$$

$$f(x) = ax + b.$$

$$\overbrace{a, b}$$

1). $a=0, b=\{z \mid z \in \mathbb{Z}\} \times$.

$$f(x) = b.$$

$b=1$. : $f(x) = 1 \Rightarrow \text{constant}$

2). $a=\{-1, 1\}, b=\{z \mid z \in \mathbb{Z}\}$

$a=1$; $f(x) = x + b.$

$b=1$;

$f_1(x) = x + 1.$

$b=-1$;

$f_{-1}(x) = x - 1$

$b=2$;

$f_2(x) = x + 2.$

} - bijective function.

$f: \mathbb{Z} \rightarrow \mathbb{Z}$

$a=2$;

$b=0$

$f(x) = 2x \rightarrow \text{Injective (one-one)}$

$\frac{1}{2} \mathbb{Z}$

PreImage: $\textcircled{1}$
even integer

★

25 - People.

9 - Goa.

8 - Bangkok.

X - Pondi

3 - X.

$9 + 8 + X + 3 = 25.$

AQ.1.6(Q-7).

A -

B -

Total no of
Subsets of A is
56 more than
the total no of
Subsets of B

$2^m \leftarrow$ A - has m-element

$2^n \leftarrow$ B - " n - "

$$2^m = 56 + 2^n.$$

$$\Rightarrow 2^m - 2^n = 56.$$

$$\Rightarrow \underline{2^n} (\underline{2^{m-n} - 1}) = 8 \times 7.$$

$$= \underline{2^3} \times \underline{7}$$

Funda.
Theor.
of
Arith.

$$\underline{2^n} = \underline{2^3} \Rightarrow \underline{n=3}$$

$$\underline{2^{m-n} - 1 = 7}$$

$$2^{m-3} - 1 = \overset{8}{7+1}$$

$$\Rightarrow \underline{2^{m-3} = 2^3}$$

$$\Rightarrow m-3=3$$

$$\Rightarrow \underline{m=6}$$

• $\mathbb{R} \times \mathbb{R} \times \mathbb{R}.$

$$f: \underline{\mathbb{N}} \rightarrow \mathbb{N}.$$

$$f(n) = \begin{cases} \frac{n+1}{2} & \text{If } n \text{ is odd} \\ \underline{\underline{n/2}} & \text{If } n \text{ is even.} \end{cases}$$

(One-one): $f(0) = \frac{0}{2} = 0, f(1) = 1, f(2) = 1$

Domain

1

Codomain

1

2

on-to

$$P = \underline{2}, \underline{2}, \underline{3}, \underline{4}, \underline{5}$$

$$Q = *$$

$$f: \mathbb{N} \rightarrow \mathbb{N} \ni k.$$

$$0 \rightarrow 0$$

$$2 \rightarrow 1$$

$$4 \rightarrow 2$$

$$2k \rightarrow k.$$

$\Rightarrow f$ is on-to.

$$\circ \text{ Dom.}(f) = \{ \underline{x} \in \mathbb{R} \mid \underline{f(x)} \in \mathbb{R} \}$$

$$\underline{\text{Dom}(f)}$$

$$f(x) = \sqrt{3x+15} \geq$$

$$\underline{x = -10}$$

$$3x+15 \geq 0 \Rightarrow \underline{x \geq -5}$$

$$[-5, \infty).$$

$$x = \underline{3} \text{ or } x = \underline{5}$$

$$x = 3 \text{ and } x = 5$$

$$\circ \{ \pi \mid -2 \leq \underline{\pi} < \underline{0} \text{ or } \underline{4} < \underline{\pi} \leq \underline{8} \}$$

$$[-2, 0) \cup (4, 8].$$

$$\underline{\{2\}} = \underline{\{2, 2\}}, \{2\}.$$

Q5

$$\underline{\underline{[2 \ 3]}} = \{x \in \mathbb{R} \mid 2 \leq x \leq 3\}$$

$$\bullet \left(\underline{\underline{(0, 2.3] \cup (2.3, 3)}} \right) \cap (0, 2.3)$$

$$\begin{array}{c} \parallel \\ 2.3 \in \underline{\underline{(0 \ 3]}} \cap \underline{\underline{(0, 2.3)}} \end{array} \begin{array}{c} 2.3 \\ \text{---} \end{array}$$

$$(0 \ 2.3)$$

$$\bullet \left(\underline{\underline{(2, 2.5] \cup (2.5, 4)}} \right) \cap (0, 3)$$

$$0 \notin \overset{x \in}{(2, 4]} \cap (0, 3) \quad \text{---} \quad \text{---}$$

$$(2, 3) \subset [2 \ 3]$$

$$\begin{array}{c} 2 < x < 3 \\ \text{---} \\ 4 \end{array}$$

$$\begin{array}{c} 0 \leq x \leq 2 \\ \text{---} \\ 4 \end{array}$$

$$3 \cdot 4 = 12$$

$$40 \quad \underline{\underline{|M \cup S \cup E|}} = \underline{\underline{|M|}} + \underline{\underline{|S|}} + \underline{\underline{|E|}} - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|$$

$$\underline{\underline{M=20}}$$

$$S=18$$

$$E=16$$

40-students



$$M \cap S = 7$$

$$M \cap E = 3$$

$$S \cap E = 9$$

$$M \cap S \cap E = 3$$

$$\underline{\underline{20-11}}$$

$$\underline{\underline{40-11}}$$

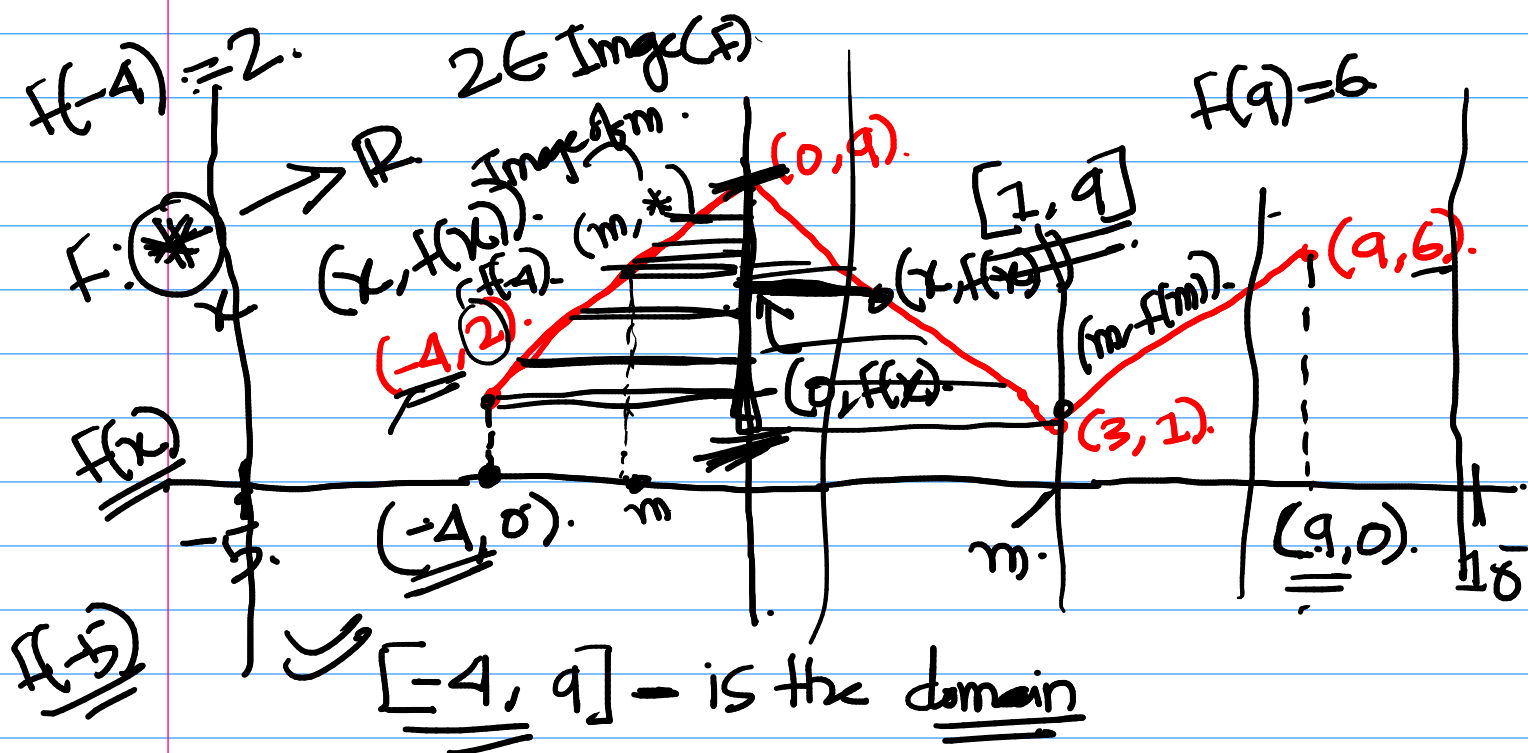
$$\underline{\text{Dom}(f)} = \{x \in \mathbb{R} \mid \underline{f(x) \in \mathbb{R}^2}\}$$

$f(x) = \sqrt{x-3} \in \mathbb{R}$

$$\begin{aligned} x-3 \geq 0 &\Rightarrow x \geq 3. \quad \text{Dom}(f) \\ &\Rightarrow x \in [3, \infty). \end{aligned}$$

$$f: [3, \infty) \rightarrow \mathbb{R}$$

$x \rightarrow +\infty$
 $x \mapsto \sqrt{x-3}$
 $3 \mapsto 0$
 $[0, \infty)$
 $\rightarrow$ Range



$$f: [-4, 9] \rightarrow \mathbb{R}$$

$$\{f(x) \mid x \in [-4, 9]\}$$

⊛ Equivalence Relation $\Leftrightarrow$ Parti. of a set.

$\underline{\underline{\{1,2\} \cup \{2,3\}}}$ ← $A = \{1,2,3\}$ $\searrow$ disjoint
 $R = \{(1,1), (2,2), (3,3)\}$
 $\checkmark A_1 = \{x \in A \mid x \sim 1\} = \{1\}$
 $\checkmark A_2 = \{x \in A \mid x \sim 2\} = \{2\}$
 $\checkmark A_3 = \{3\}$

$\{1,2\} \cup \{2,3\}$
 $\underline{\underline{\{2\} \cup \{1,3\}}}$
 $\underline{\underline{\{1\} \cup \{2\} \cup \{3\}}}$

$R = \{(1,1), (2,2), (3,3), (1,3), (3,1)\}$

$\checkmark A_1 = \{x \in A \mid x \sim 1\} = \{1,3\}$ $\{1,3\} \cup \{2\}$
 $A_2 = \{x \in A \mid x \sim 2\} = \{2\}$

~~$A_3 = \{x \in A \mid x \sim 3\} = \{1,3\}$~~

$a \in A_3 \quad a \sim 3. \quad 3 \sim 1 \Rightarrow a \sim 1.$

$\Rightarrow \underline{\underline{a \in A_1}}$

⊛ $(a,a) \forall a \in A$

★ R on A.

$$\text{If } (a,b) \in R \text{ \& } (b,a) \in R \Rightarrow a=b$$

$$A = \{1, 2, 3, 4\}$$

$$R = \{(\underline{1}, 2), (3, 3), (2, 1), (\underline{1}, 4)\}$$

$$(\underline{1}, 2) \in R \text{ \& } (2, 1) \in R \text{ But } \underline{1} \neq \underline{2}$$

R-is not anti-sy. relation.

$$(\underline{1}, 1) \in R \text{ \& } (1, 1) \in R$$

$$\checkmark R = \{(\underline{1}, 1), (\underline{1}, 4), (\underline{3}, 1)\}$$

$$\begin{matrix} a & b. \\ (1, 1) \in R \text{ \& } (1, 1) \in R & a=b \end{matrix}$$

$$\checkmark (\underline{1}, 4) \in R$$

$$(3, 1) \in R$$

$$\star \text{ If } (a,b) \in R \Rightarrow \text{asy } (b,a) \notin R.$$

$$f: \mathbb{N} \rightarrow \mathbb{N}.$$

$$\underline{f(x)} = \underline{|(x-4)(4x-10)|} \geq \underline{0}$$

$$\underline{f(x)=0}$$

$$\underline{f(4)=0}$$

$$\underline{f(10/4)=0}$$

$$\underline{2.5}$$

$$A = \{0, \textcircled{2}, 4, 6, 8, \dots\}$$

C 3 & A
Not even

A = all the even N.

- B = all odd nat. N.

$$C = n/100$$

$$D =$$

$$P < 100$$

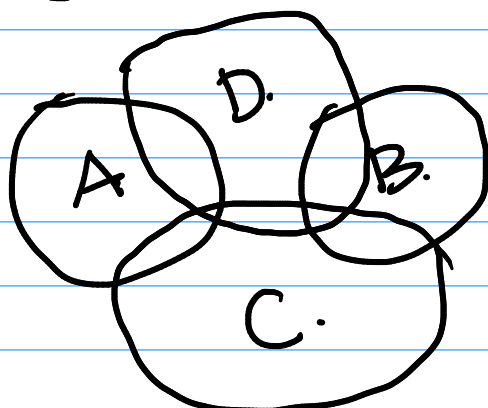
$$m/n$$

$$n = m \times k$$

① $\cup B = \{1, \check{3}, \check{5}, \check{7}, \dots\}$ $m = 0/100 = n$
 $100 = \underline{\underline{0 \times k}}$

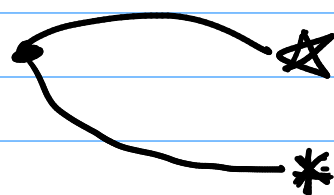
$$C = \{1, 2, 4, 5, 10, \dots, 100\}$$

$$D = \{\textcircled{2}, \check{3}, \check{5}, \check{7}, 11, 13, 17, 19, \dots\}$$



$\cup \{x_1 = x_2\}$
 $f(x_1) = f(x_2)$

It $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$



○ $A = \{K, D, H\}$ R(A \times A)

$\cup R = \{(\underline{\underline{K}}, \underline{\underline{K}}), (K, D), (D, K), (D, D), (H, H)\}$

anb

$$\underline{\underline{2}} \in \left\{ n \in \mathbb{N} \mid \begin{array}{l} 1 \in \mathbb{N}, \sqrt{1} \in \mathbb{N} \\ n \in \mathbb{N}, \sqrt{n} \in \mathbb{N}, \\ 1 < \underline{\underline{n}} < 10000 \end{array} \right\}$$

$$\sqrt{1} = 1. \quad \{1, 4, 9, 16, \dots, 99^2\}$$

$$\underline{\underline{100}}^2$$

- min. $|x| \geq 0$

$$f(x) = 9|x| - 8$$

$$\underline{\underline{x \geq 0}} \quad \underline{\underline{-8}}$$

- max $x = -8$

$$f(x) = 11 - \underline{\underline{|x+8|}}$$

$$\underline{\underline{11}} - \underline{\underline{\text{max}}}$$